西湖大學 WESTLAKE UNIVERSITY

Bioinformatics training – 2

高通量测序数据简介及质量控制

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Dec 9th, 2021

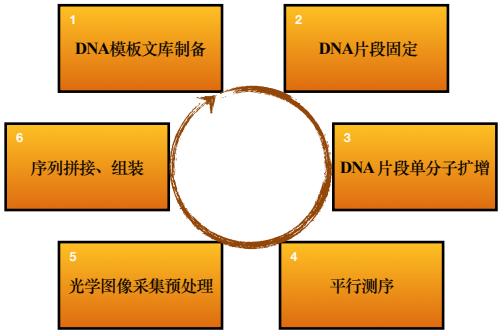
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- 高通量测序技术结果格式
- 高通量测序文件质量控制
- 高通量测序文件数据库

第二代测序技术

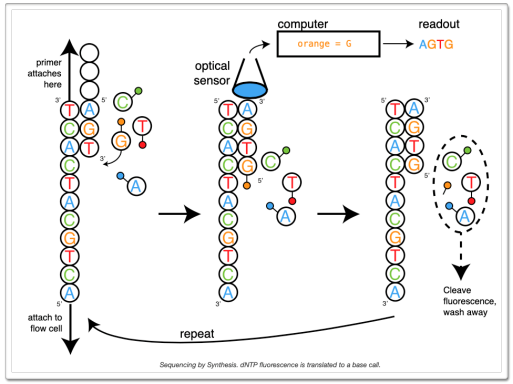
- 第二代测序技术又称高通量测序技术，以能一次并行对几十万到几百万条DNA分子进行序列测定和一般读长较短等为标志。目前常用的高通量测序平台主要有来自罗氏公司的 454平台、Illumina公司Illumina测序仪和 ABI 公司的 SOLiD 测序仪。

第二代测序技术基本流程



Illumina测序仪的原理与流程

1. DNA文库制备
2. Flowcell结合
3. 桥式PCR与变性
4. 测序



测序结果如何存储?



你的猫.gif



你的猫.jpg



你的猫.png



FASTQ文件格式

- FASTQ format is a text-based format for storing both a nucleotide sequence and its corresponding quality scores.
  - Both the sequence letter and quality score are encoded with a single ASCII character.
  - It was originally developed at the Wellcome Trust Sanger Institute to bundle a FASTA sequence and its quality data.
  - It has recently become the standard for storing the output of high throughput sequencing instruments such as the Illumina Genome Analyzer.

FASTA文件格式

- FASTA format is a simple type of format that bioinformaticians use to represent either nucleotide or protein sequences. It is written in text format, allowing for processing tools to easily parse the data. The general file extension is .fa.
- The FASTA file format originated from a DNA and protein sequence alignment software package created in the mid-1980's. The format allows you to precede each sequence with a comment.
- There are two lines per sequence - 1) the identifier (comments, annotations) and 2) the sequence itself.

FASTA文件格式



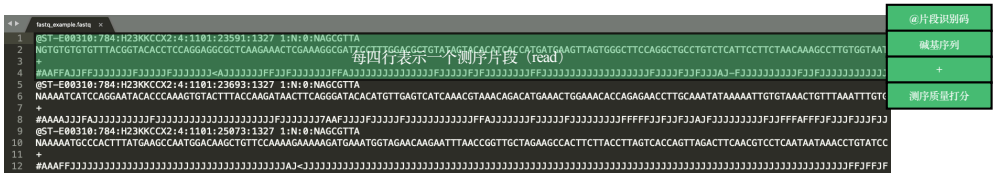
Identifier

Sequence

FASTQ文件格式

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FASTQ文件格式

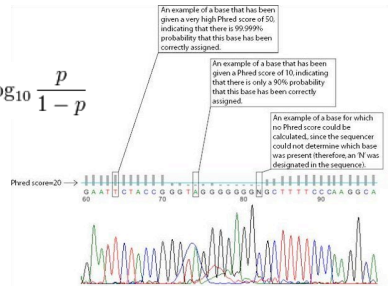

fastq_example.fastq
@ST-E00310:784:H23KKCCX2:4:1101:23591:1327 1:N:0:NAGCGTTA
NGTGTGTGTTTACGGTACACCTCCAGGAGGCGCTCAAGAAACTCGAAAGGCGATTCCTTTGGAGCGTGTAAGTACACATCACATGATGAAGTTAGTGGGCTTCCAGGCTGCCTGTCTCATTCCTTCTAACAAAGCCTTGTGGTAAT
+
#AAFFAJJFFJJJJJJJFJJJJJFJJJJJJJ<AJJJJJJJFFJJFJJJJJJJFFAJJJJJJJJJJJJJJJFJJJJFJFJJJJJJJJFFJJJJJJJJJJJJJJJJJJJJFJJJJFJJFJJJAJ-FJJJJJJJJJJFJJFJJJJJJJJJJJ
@ST-E00310:784:H23KKCCX2:4:1101:23693:1327 1:N:0:NAGCGTTA
NAAAATCATCCAGGAATACACCCAAAGTGTACTTTACCAAGATAACTTCAGGGATACACATGTTGAGTCATCAAACGTAAACAGACATGAAACTGGAAACACCAGAGAACCTTGCAAATATAAAAATTGTGTAAACTGTTTAAATTTGTC
+
#AAAAJJJFAJJJJJJJJJFJJJJJJJJJJJJJJJJJJJJJJJJFJJJJJJJ7AAFJJJJFJJJJJFJJJJJJJJJJJJFFAJJJJJJJFJJJJJFJJJJJJJJJJFFFFFJJFJJFJJAJFJJJJJJJJJFJJFFFAFFFJFJJJFJJJFJJ
@ST-E00310:784:H23KKCCX2:4:1101:25073:1327 1:N:0:NAGCGTTA
NAAAAATGCCCACTTTATGAAGCCAATGGACAAGCTGTTCCAAAAGAAAAAGATGAAATGGTAGAACAAGAATTTAACCGGTTGCTAGAAGCCACTTCTTACCTTAGTCACCAGTTAGACTTCAACGTCCTCAATAATAAACCTGTATCC
+
#AAAFFJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJAJ

测序质量Phred打分

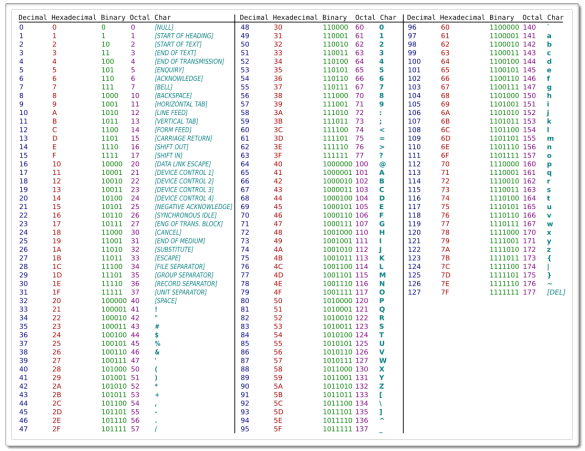
- Phred quality scores were originally developed by the program Phred to help in the automation of DNA sequencing in the Human Genome Project.
- A quality value *Q* is an integer mapping of *p* (i.e., the probability that the corresponding base call is incorrect).

$$Q_{\text{sanger}} = -10 \log_{10} p$$

$$Q_{\text{solexa-prior to v.1.3}} = -10 \log_{10} \frac{p}{1-p}$$



ASCII对照表



Decimal	Hexadecimal	Binary	Octal	Char
0	0	0	0	[NULL]
1	1	1	1	[START OF HEADING]
2	2	10	2	[START OF TEXT]
3	3	11	3	[END OF TEXT]
4	4	100	4	[END OF TRANSMISSION]
5	5	101	5	[ENQUIRY]
6	6	110	6	[ACKNOWLEDGE]
7	7	111	7	[BELL]
8	8	1000	10	[BACKSPACE]
9	9	1001	11	[HORIZONTAL TAB]
10	A	1010	12	[LINE FEED]
11	B	1011	13	[VERTICAL TAB]
12	C	1100	14	[FORM FEED]
13	D	1101	15	[CARRIAGE RETURN]
14	E	1110	16	[SHIFT OUT]
15	F	1111	17	[SHIFT IN]
16	10	10000	20	[DATA LINK ESCAPE]
17	11	10001	21	[DEVICE CONTROL 1]
18	12	10010	22	[DEVICE CONTROL 2]
19	13	10011	23	[DEVICE CONTROL 3]
20	14	10100	24	[DEVICE CONTROL 4]
21	15	10101	25	[NEGATIVE ACKNOWLEDGE]
22	16	10110	26	[SYNCHRONOUS IDLE]
23	17	10111	27	[END OF TRANS. BLOCK]
24	18	11000	30	[CANCEL]
25	19	11001	31	[END OF MEDIUM]
26	1A	11010	32	[SUBSTITUTE]
27	1B	11011	33	[ESCAPE]
28	1C	11100	34	[FILE SEPARATOR]
29	1D	11101	35	[GROUP SEPARATOR]
30	1E	11110	36	[RECORD SEPARATOR]
31	1F	11111	37	[UNIT SEPARATOR]
32	20	100000	40	[SPACE]
33	21	100001	41	!
34	22	100010	42	"
35	23	100011	43	#
36	24	100100	44	$
37	25	100101	45	%
38	26	100110	46	&
39	27	100111	47	'
40	28	101000	50	(
41	29	101001	51	)
42	2A	101010	52	*
43	2B	101011	53	+
44	2C	101100	54	,
45	2D	101101	55	-
46	2E	101110	56	.
47	2F	101111	57	/
48	30	110000	60	0
49	31	110001	61	1
50	32	110010	62	2
51	33	110011	63	3
52	34	110100	64	4
53	35	110101	65	5
54	36	110110	66	6
55	37	110111	67	7
56	38	111000	70	8
57	39	111001	71	9
58	3A	111010	72	:
59	3B	111011	73	;
60	3C	111100	74	<
61	3D	111101	75	=
62	3E	111110	76	>
63	3F	111111	77	?
64	40	1000000	100	@
65	41	1000001	101	A
66	42	1000010	102	B
67	43	1000011	103	C
68	44	1000100	104	D
69	45	1000101	105	E
70	46	1000110	106	F
71	47	1000111	107	G
72	48	1001000	110	H
73	49	1001001	111	I
74	4A	1001010	112	J
75	4B	1001011	113	K
76	4C	1001100	114	L
77	4D	1001101	115	M
78	4E	1001110	116	N
79	4F	1001111	117	O
80	50	1010000	120	P
81	51	1010001	121	Q
82	52	1010010	122	R
83	53	1010011	123	S
84	54	1010100	124	T
85	55	1010101	125	U
86	56	1010110	126	V
87	57	1010111	127	W
88	58	1011000	130	X
89	59	1011001	131	Y
90	5A	1011010	132	Z
91	5B	1011011	133	[
92	5C	1011100	134	\
93	5D	1011101	135	]
94	5E	1011110	136	^
95	5F	1011111	137	_
96	60	1100000	140	`
97	61	1100001	141	a
98	62	1100010	142	b
99	63	1100011	143	c
100	64	1100100	144	d
101	65	1100101	145	e
102	66	1100110	146	f
103	67	1100111	147	g
104	68	1101000	150	h
105	69	1101001	151	i
106	6A	1101010	152	j
107	6B	1101011	153	k
108	6C	1101100	154	l
109	6D	1101101	155	m
110	6E	1101110	156	n
111	6F	1101111	157	o
112	70	1110000	160	p
113	71	1110001	161	q
114	72	1110010	162	r
115	73	1110011	163	s
116	74	1110100	164	t
117	75	1110101	165	u
118	76	1110110	166	v
119	77	1110111	167	w
120	78	1111000	170	x
121	79	1111001	171	y
122	7A	1111010	172	z
123	7B	1111011	173	{
124	7C	1111100	174	|
125	7D	1111101	175	}
126	7E	1111110	176	~
127	7F	1111111	177	[DEL]

几种测序质量打分版本

- Beware of different versions:

```
  SSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSS.....................................................
  ..........................XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX.....................
  ............................IIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIII.....................
  ..............................JJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJ.....................
  LLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLL.....................................................
  !"#$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~
  |                         |    |        |                              |                     |
33                        59   64       73                             104                   126

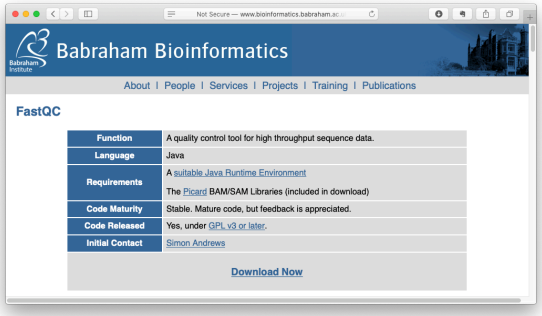
S - Sanger        Phred+33,  raw reads typically (0, 40)
X - Solexa        Solexa+64, raw reads typically (-5, 40)
I - Illumina 1.3+ Phred+64,  raw reads typically (0, 40)
J - Illumina 1.5+ Phred+64,  raw reads typically (3, 40)
   with 0=unused, 1=unused, 2=Read Segment Quality Control Indicator (bold)
L - Illumina 1.8+ Phred+33,  raw reads typically (0, 41)
```

- With sequence data recently produced, you do not need to worry.
- For older data, be careful.

测序文件质量控制

- 高通量测序技术可在一次实验中生成数以千万计个序列片段。在分析这些序列片段并得出生物学结论之前，我们应该始终对这些序列片段进行质量控制检查，从而确保这些原始数据质量可靠，并且没有任何可能影响下游分析的问题或者误差。
- FastQC： http://www.bioinformatics.babraham.ac.uk/projects/fastqc/

FastQC—针对高通量测序数据的质量控制工具



FastQC的安装、使用和结果

FastQC A quality control application for high throughput sequence data

- README
- Installation and setup instructions
- Release Notes Please read these before using the program.
- FastQC v0.11.9 (Win/Linux zip file)
- FastQC v0.11.9 (Mac DMG image)
- Source Code for the latest FastQC release

使用命令行:

```
(base) zhaochengchen@PeiLab:~$ fastqc --help

            FastQC - A high throughput sequence QC analysis tool

SYNOPSIS

        fastqc seqfile1 seqfile2 .. seqfileN

        fastqc [-o output dir] [--(no)extract] [-f fastq|bam|sam]
               [-c contaminant file] seqfile1 .. seqfileN

DESCRIPTION

    FastQC reads a set of sequence files and produces from each one a quality
    control report consisting of a number of different modules, each one of
    which will help to identify a different potential type of problem in your
    data.

    If no files to process are specified on the command line then the program
    will start as an interactive graphical application.  If files are provided
    on the command line then the program will run with no user interaction
    required.  In this mode it is suitable for inclusion into a standardised
    analysis pipeline.

    The options for the program as as follows:

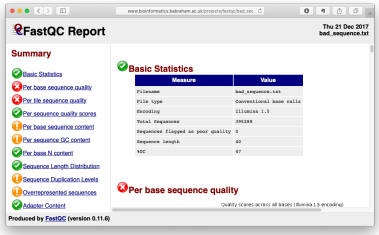
    -h --help       Print this help file and exit

    -v --version    Print the version of the program and exit

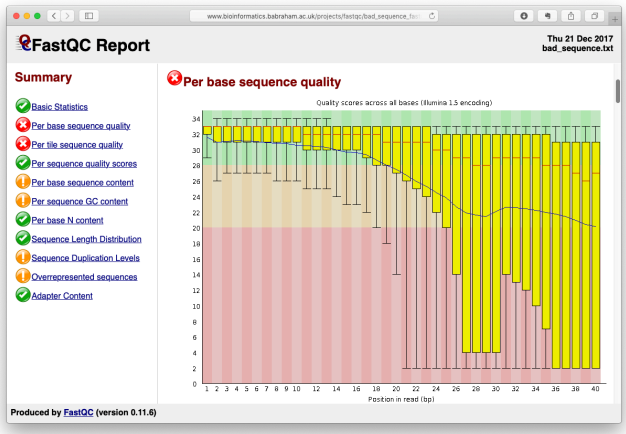
    -o --outdir     Create all output files in the specified output directory.
                    Please note that this directory must exist as the program
                    will not create it.  If this option is not set then the
                    output file for each sequence file is created in the same
                    directory as the sequence file which was processed.
```

结果:

your_fastq_file_name_fastqc.html
your_fastq_file_name_fastqc.zip

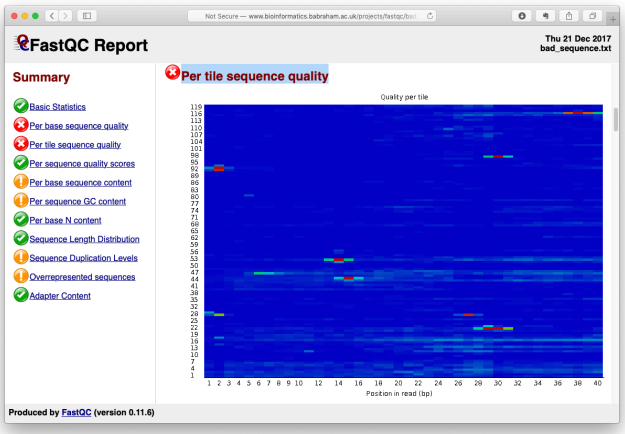


FastQC—各个位置的碱基质量值分布



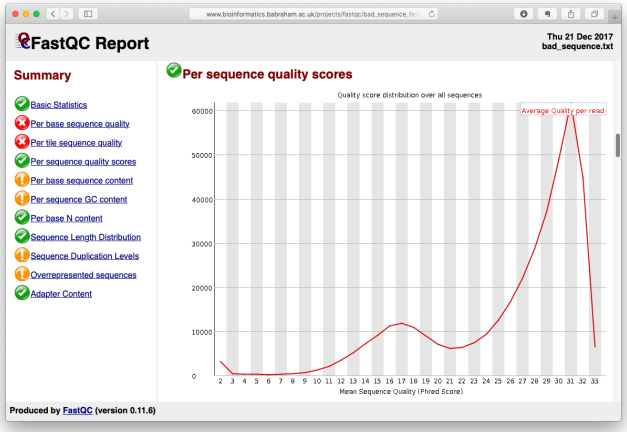
- 横坐标代表碱基在测序片段上的位置，纵坐标表示每一个位置所对应的测序质量值。
- 红色表示中位数，黄色是25%~75%区间，短横线是10%~90%区间，蓝线是平均数。

FastQC—各个tile的碱基质量值分布



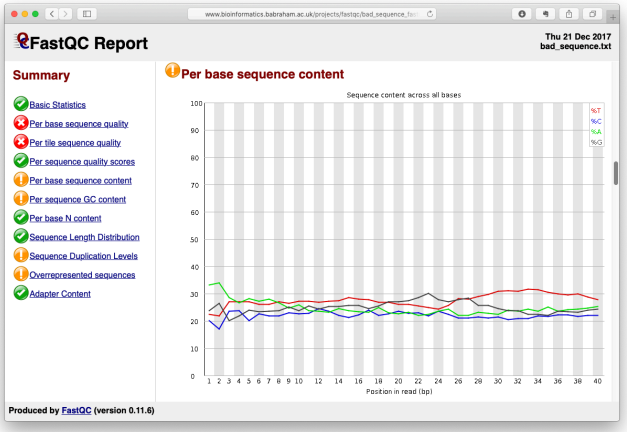
- 横坐标代表 tile在测序片段上的位置，纵坐标表示每一个tile对应的Index编号。
- 蓝色代表测序质量很高，暖色代表测序质量不高，如果某些位置出现暖色，可以在后续分析中把该位置测序的结果全部都去除。

FastQC—碱基的总体质量值分布



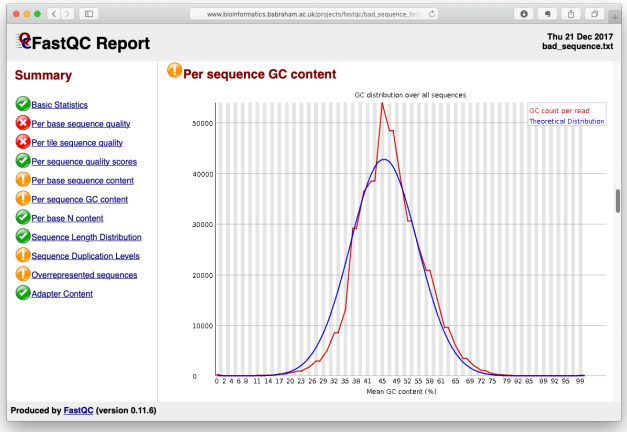
- 横坐标表示片段测序质量的均值；纵坐标代表片段个数。
- 用来检测是否有部分测序片段质量较低。

FastQC—各个位置上碱基比例分布



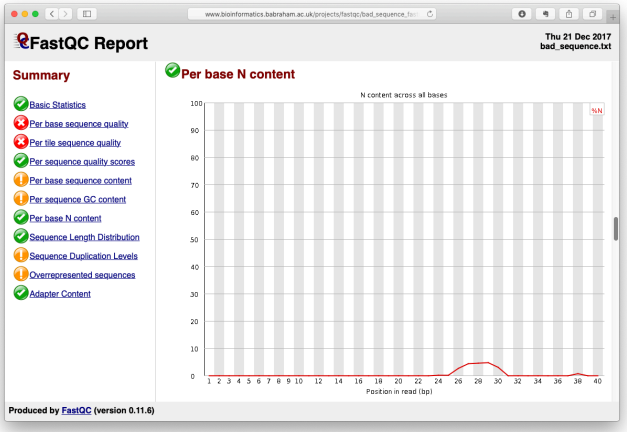
- 横坐标代表碱基在测序片段上的位置；纵坐标代表ATCG四种碱基在该位置上的百分比。
- 用于分析碱基的分离程度。

FastQC—GC含量分布图



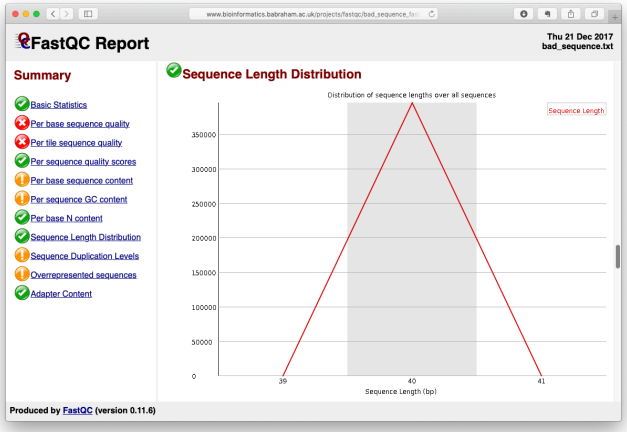
- 横坐标表示GC含量；纵坐标表示对应该GC含量的序列片段数目。
- 用于分析GC含量的正常与否。

FastQC—N含量分布图



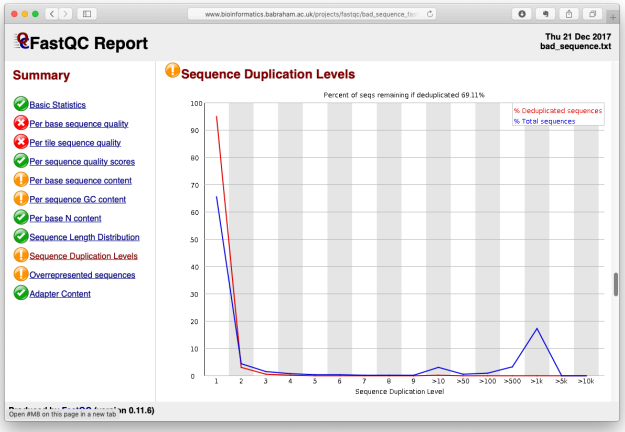
- 横坐标表示碱基在测序片段上的位置；纵坐标代表N在该位置上的百分比。
- N在测序数据中一般是不应该出现的，如果出现则意味着，测序的光学信号无法被清晰分辨，如果这种情况多的话，往往意味着测序系统或者测序试剂的错误。

FastQC—测序长度分布



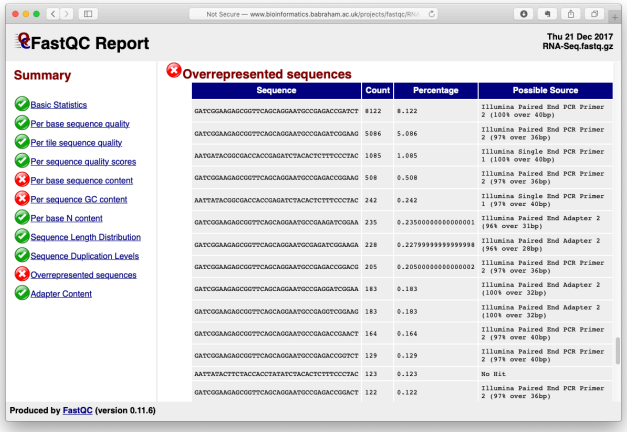
- 横坐标表示测序片段的长度；纵坐标表示对应该片段长度的序列片段数目。

FastQC—测序片段重复率



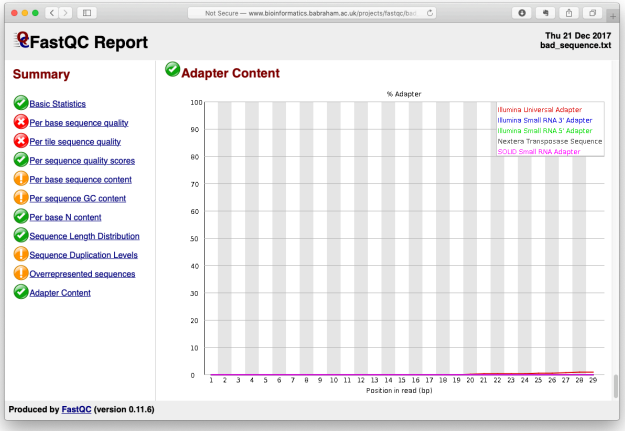
- 横坐标表示测序片段的重复程度；纵坐标表示对应重复程度下的测序片段比例。

FastQC—测序片段共有序列



- 该表格列出了数据中部分测序片段共有的序列比例以及可能的来源。
- 用于分析测序片段中是否还存在接头序列。

FastQC—接头序列含量

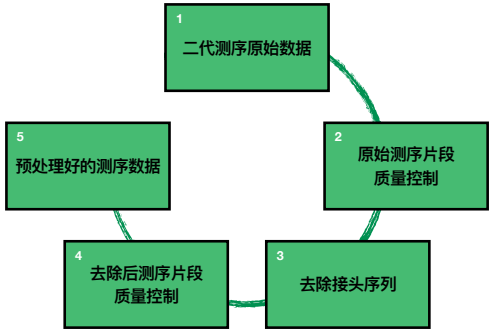


- 横坐标代表碱基在测序片段上的位置，纵坐标代表该位置接头序列含量的百分比。

利用cutadapt去除接头序列

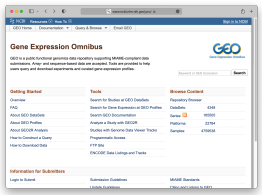


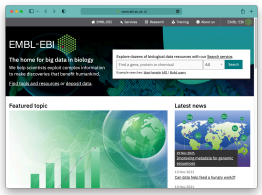
小结

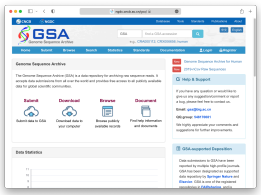


高通量数据库

- 为了便于测序数据的发布和共享，高通量测序数据以 FASTQ 格式来记录所测的碱基读段和质量分数。 NCBI、EBI、GSA 等数据中心建立了大容量的数据库来存放共享的测序数据。

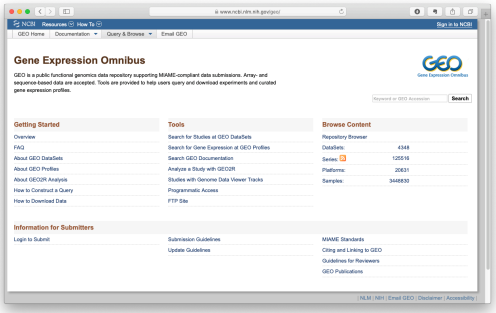






GEO数据库

- Gene Expression Omnibus (GEO)
  - http://www.ncbi.nlm.nih.gov/geo/
  - 一个公共存储库，用于存档和发布研究人员提交的基因表达数据。
  - 在线资源，用于基因表达数据浏览、查询、分析和检索。
  - 根据资助机构和期刊的要求，方便存放基因表达数据。

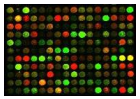


GEO数据库的结构

- GEO数据库中包含了四种数据记录类别。
  - Platform (GPL)：实验平台，记录获得数据的实验技术。
  - Sample (GSM)：单个实验样本，记录一次实验中单个样本的具体信息。
  - Series (GSE)：一次实验数据集，记录一次实验中实验的设计及所有相关的实验样本的信息。
  - DataSets (GDS)：数据集合。由 GEO 工作人员负责整理的样本数据集合。

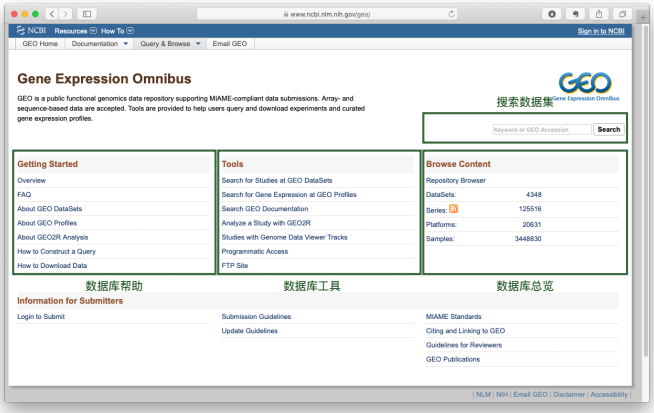
GEO数据库的结构

Submitted by Manufacturer	Submitted by Experimentalists		Curated by NCBI
GPL	GSM	GSE	GDS
Platform descriptions	Raw/processed spot intensities from a single slide/chip	Grouping of slide/ chip data "a single experiment"	Grouping of experiments

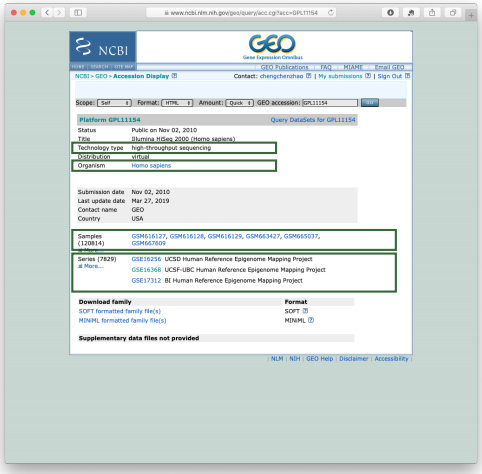





GEO—首页



GEO—GPL 页面

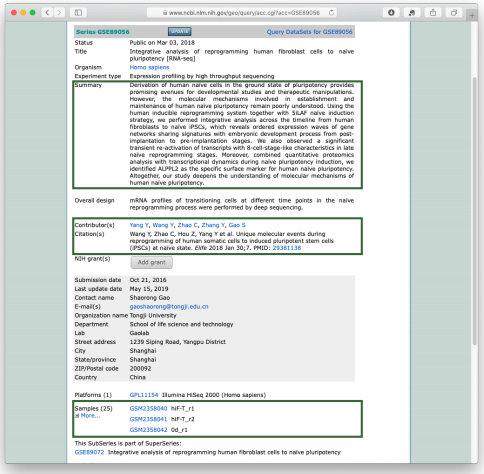


产生数据的技术手段
针对的物种

采用该平台的具体的实验样本

采用该平台的具体的实验数据集

GEO—GSE 页面

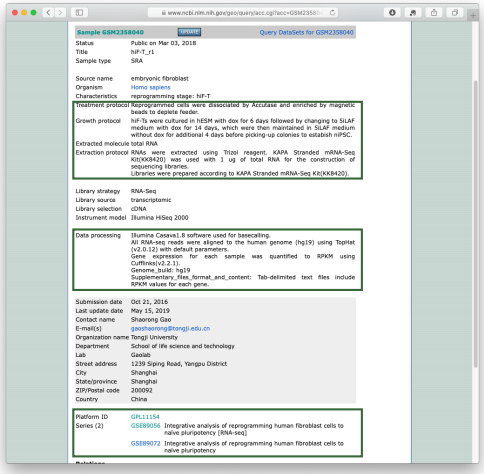


该次实验的设计思路及包含的样本信息总述。

该次实验的贡献者及相关文献。

该次实验中包含的各个样本（sample）详细情况。

GEO—GSM 页面



得到本次样本时所用的具体实验条件。

得到本次样本数据时所用的具体分析条件。

本次样本所属的GSE 号和 GPL号。

在服务器上下载GEO数据

- DataDownloadFromGEO_wget.py -i GSMID -o OutputName
  - -i: input the GSM id
  - -o: output file name
- After the program runs successfully you will get a file named OutputName.sra.

使用fastq-dump将sra文件转化为fastq文件

- Single-end
- fastq-dump --gzip OutputName.sra
- After the program runs successfully you will get a file named OutputName.fastq.gz.

- Pair-end:
- fastq-dump --gzip --split-files OutputName.sra
- After the program runs successfully you will get two files named OutputName_1.fastq.gz & OutputName_2.fastq.gz.

Exercise 4

1. 这是上周的一篇nature文章：https://www.nature.com/articles/s41586-021-04267-8，该文章将其产生的高通量测序数据上传到了GEO数据库中，从文章中找到GSE号，并在GEO中打开页面。
2. 进入该数据集的子集（Human blastoids model blastocyst development and implantation [Bulk RNA-seq]）中，选择160505-Naive_re_derived_C1这套数据，得到其GSM号。
3. 下载其原始测序文件，并转换成fastq格式。
4. 利用fastQC对该测序文件进行测序质量分析。

Exercise 4-2

1. 学有余力的同学，请尝试重现fastQC的几张结果图。
   - Per base sequence quality
     - boxplot()
     - points()
   - Per tile sequence quality
     - image()
   - Per sequence quality scores
     - mean()
     - plot()
   - Per base sequence content
     - plot()